2023.M24

2023L005A1EL

Coimisiún na Scrúduithe Stáit
State Examinations Commission

Leaving Certificate Examination 2023

Geography

Part One and Answerbook

Higher Level

Friday 9 June Morning 9:30 - 12:20

400 marks

Examination Number

Day and Month of Birth

For example, 3rd February is entered as 0302

Centre Stamp

Instructions

Write your Examination Number and your Day and Month of Birth in the boxes on the front cover.

Write your answers to all parts of the examination into this answerbook. This answerbook will be scanned and your work will be presented to an examiner on screen. Anything that you write outside of the answer areas may not be seen by the examiner.

Write your answers in blue or black pen. You may use pencil for sketches, graphs and diagrams only.

There are two parts to this examination. Questions for **PART TWO** are supplied separately but your answers must be written in this answerbook.

PART ONE Short-Answer Questions **80 marks**

Attempt any **TEN** questions.

All questions carry equal marks.

PART TWO Structured and Essay Questions **320 marks**

You must attempt **FOUR** questions. This must include one question from three of the sections below:

- Section 1: **Patterns and Processes in the Physical Environment** (Questions 1-3)
- Section 2: **Regional Geography** (Questions 4-6)
- Section 3: **Electives** (Questions 7-12)
- Section 4: **Options** (Questions 13-24)

The fourth question can be from any section.

All questions carry equal marks.

PART ONE – SHORT-ANSWER QUESTIONS – 80 MARKS

Attempt any **TEN** questions.

All questions carry equal marks.

THESE QUESTIONS ARE TO BE ANSWERED IN THE SPACES PROVIDED

1. Landscape Identification

Examine the 1:50 000 Ordnance Survey map and legend accompanying this paper.

(i) In the table below, match each of the following landscape features with the grid reference that best matches its location on the Ordnance Survey map:

- Drumlin
- Concave slope
- Confluence.

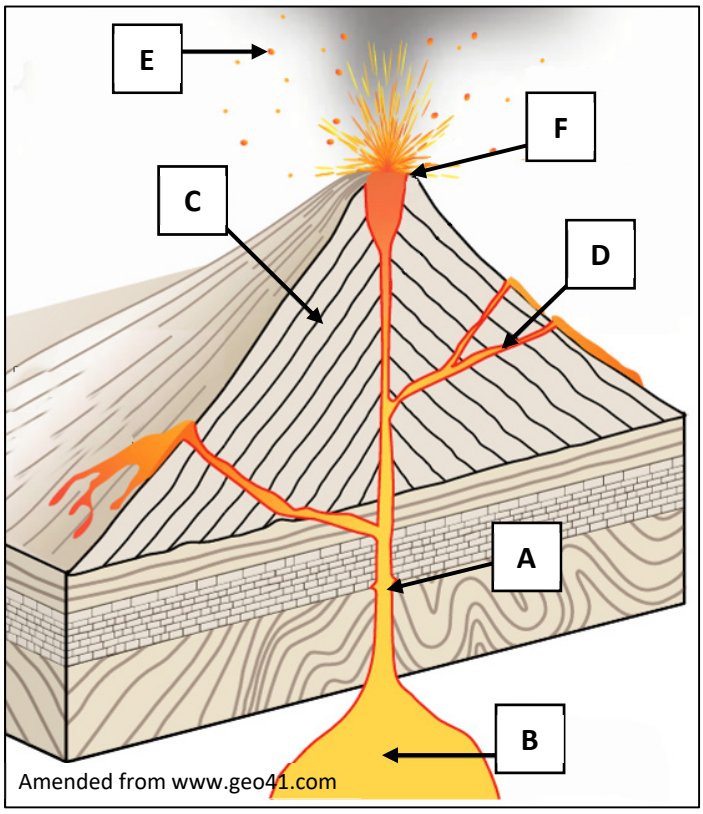
Grid Reference	Feature
H 50 05	
H 39 01	
H 45 01	

(ii) Name **two** lakes found on the map that are north of Northing 03.

1.
2.

[8m]

2. Volcanic Activity



Examine the diagram of an erupting volcano above and answer each of the following questions.

(i) Match each of the letters **A**, **B**, **C**, **D**, **E** and **F** with the term that best matches it in the table below.

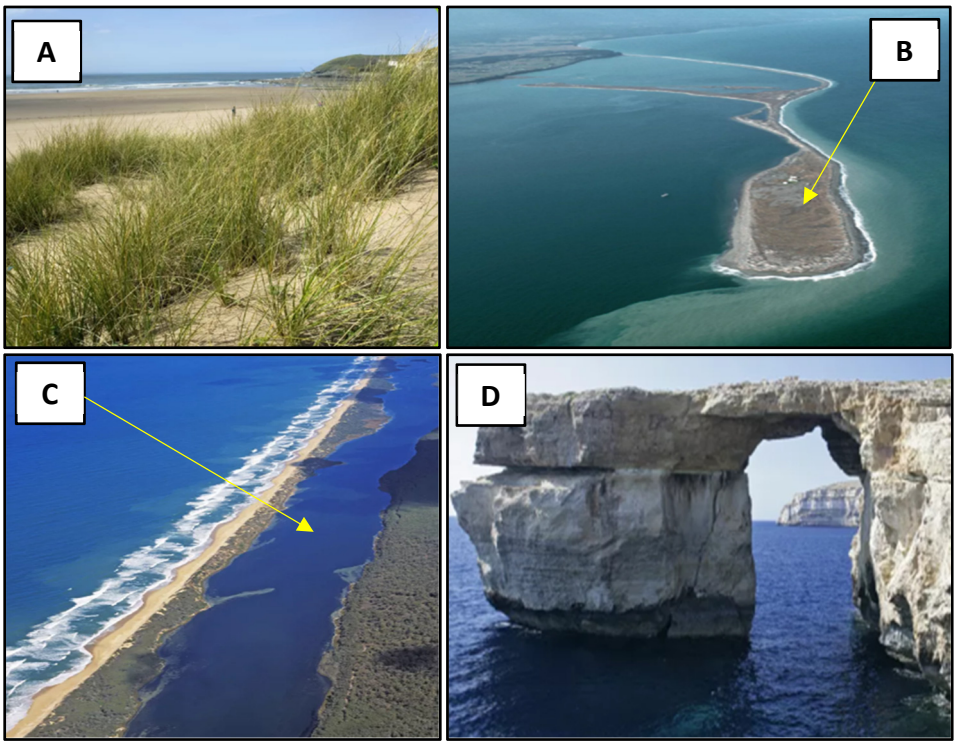
Term	Letter
Layers of lava and volcanic ash	
Crater	
Secondary vent	
Magma chamber	
Main vent	
Volcanic bomb	

(ii) Name **two** plutonic features that form when magma cools.

1.
2.

[8m]

3. Coastal Landscape



Amended from www.dkfindout.com

Examine the photographs above and answer each of the following questions.

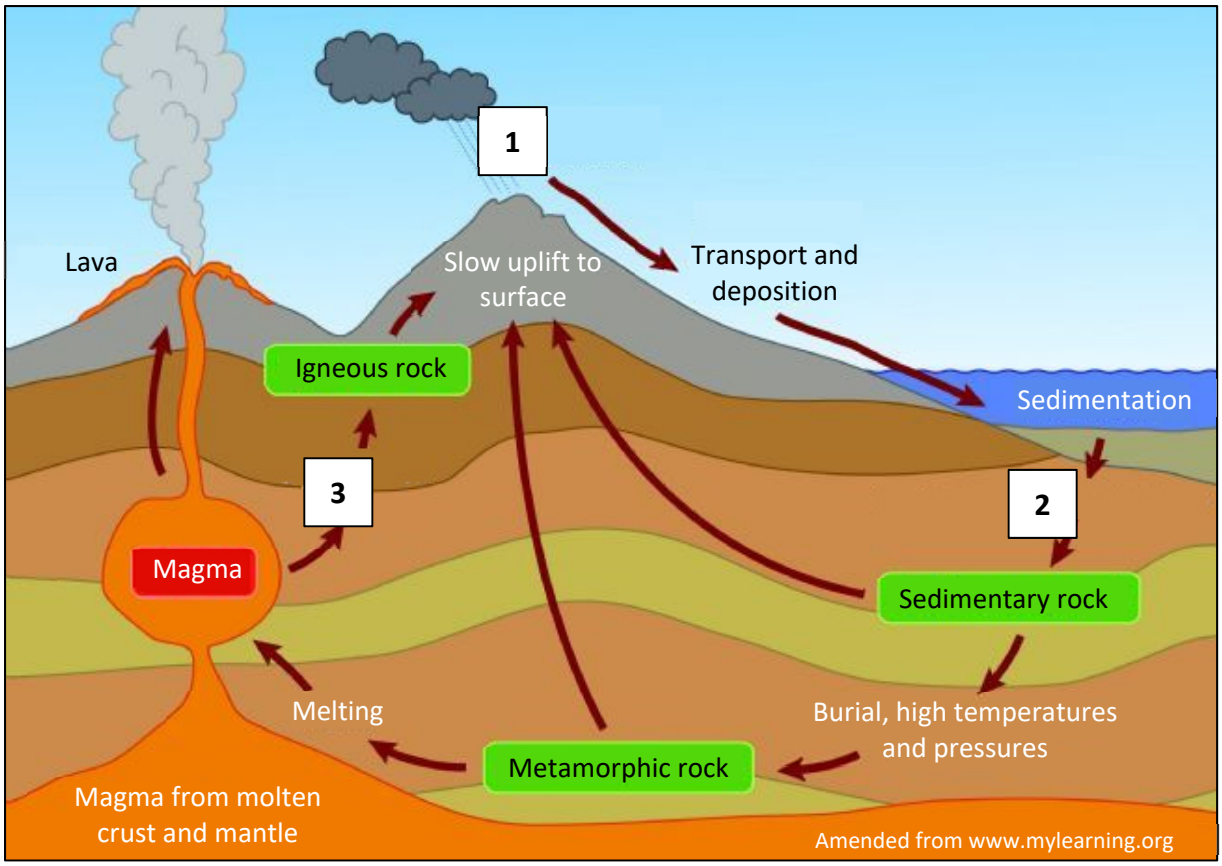
(i) Match each of the photographs **A**, **B**, **C** and **D** with the feature that best matches it in the table below.

(ii) Indicate, by ticking the correct box, whether each of the features in the table below, is most associated with processes of erosion or the process of deposition.

Q.3(i)	**Feature**	**Q.3(ii)**	
Letter		**Processes of Erosion** (✓)	**Process of Deposition** (✓)
	Lagoon		
	Sand dunes		
	Sea arch		
	Sand spit		

[8m]

4. Rocks



Examine the diagram of the rock cycle above and answer each of the following questions.

(i) Match each of the numbers on the diagram with the process that best matches it in the table below.

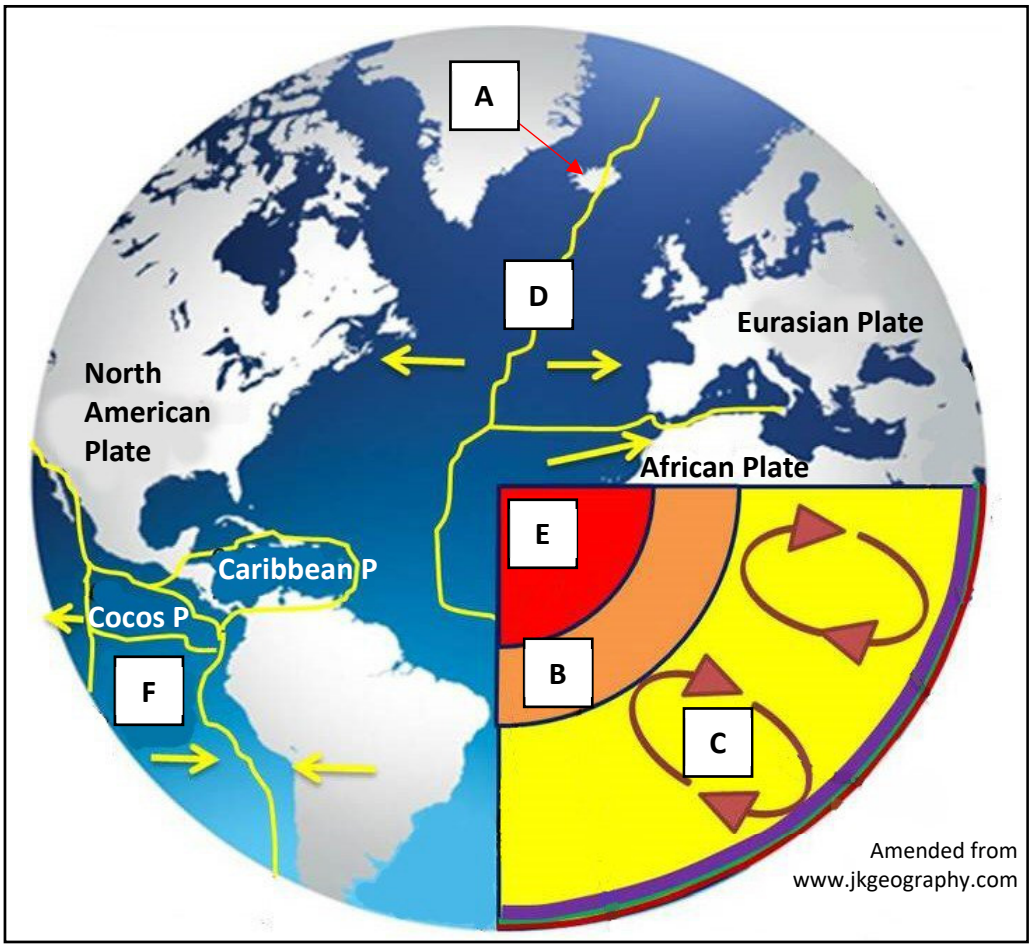
Process	Number
Crystallisation of magma	
Weathering and erosion	
Compaction and cementation	

(ii) Name **one** example of a sedimentary rock **and one** example of a metamorphic rock.

Sedimentary rock:
Metamorphic rock:

[8m]

5. Plate Tectonics



Examine the diagram of the earth above and answer each of the following questions.

(i) Match each of the letters **A**, **B**, **C**, **D**, **E** and **F** with the term that best matches it in the table below.

Term	Letter
Convection currents	
Inner core	
Outer core	
Nazca plate	
Volcanic island	
Divergent plate boundary	

(ii) What is the name for the solid layer of rock around the earth which consists of the upper mantle and the crust?

[8m]

6. Concept of a Region

(i) Complete the table below by matching each of the examples of regions in the list, with the type of region most associated with it in the table.

Examples of Regions:

West of Ireland	North European Plain
Cool Temperate Oceanic region	London
Sambre Meuse Valley	County Meath

Type of Region	Example of Region
Urban region	
Peripheral region	
Climatic region	
Geomorphological region	
Administrative region	
Region of industrial decline	

(ii) Indicate whether each of the following statements is true or false by ticking (✓) the correct box.

(a) Core regions are usually well-developed socio-economic regions that attract industry.

True ☐ False ☐

(b) Peripheral regions experience net immigration.

True ☐ False ☐

[8m]

7. Map Skills

Examine the 1:50 000 Ordnance Survey map and legend accompanying this paper, and answer each of the following questions.

(i) What is the height (in metres) at the triangulation pillar at H 462 008?

(ii) Measure the straight-line distance (in kilometres) from the centre of the roundabout at H 423 019 to the centre of the roundabout at H 435 041.

(iii) Give a six-figure grid reference for the exact point where the N3 exits the map in the southeast.

(iv) Calculate the area (in km^2) of the map extract which is north of Northing 00.

[8m]

8. Aerial Photograph and Ordnance Survey Map

Examine the aerial photograph of Cavan and the 1:50 000 Ordnance Survey map and legend accompanying this paper and answer each of the following questions.

(i) In what direction was the camera pointed when the aerial photograph was taken?

(ii) Give a four-figure grid reference for the lake that is located in the right foreground of the aerial photograph.

(iii) Which of the following regional roads can be seen in the left background of the aerial photograph? Tick (✓) the correct box.

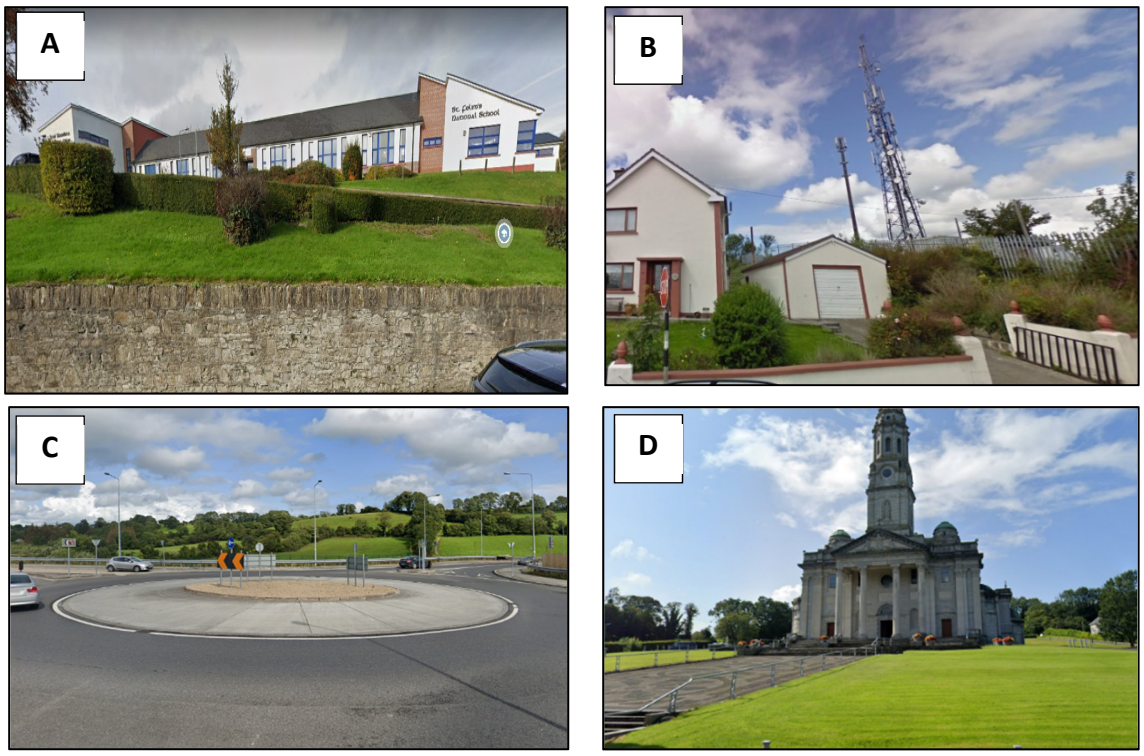
R 212 ☐ R 188 ☐ R 198 ☐ R 165 ☐

(iv) State the **type** of aerial photography used for the aerial photograph accompanying this paper.

[8m]

9. Aerial Photograph

Examine the aerial photograph of Cavan accompanying this paper and the Google Street View images below from the same area of Cavan.



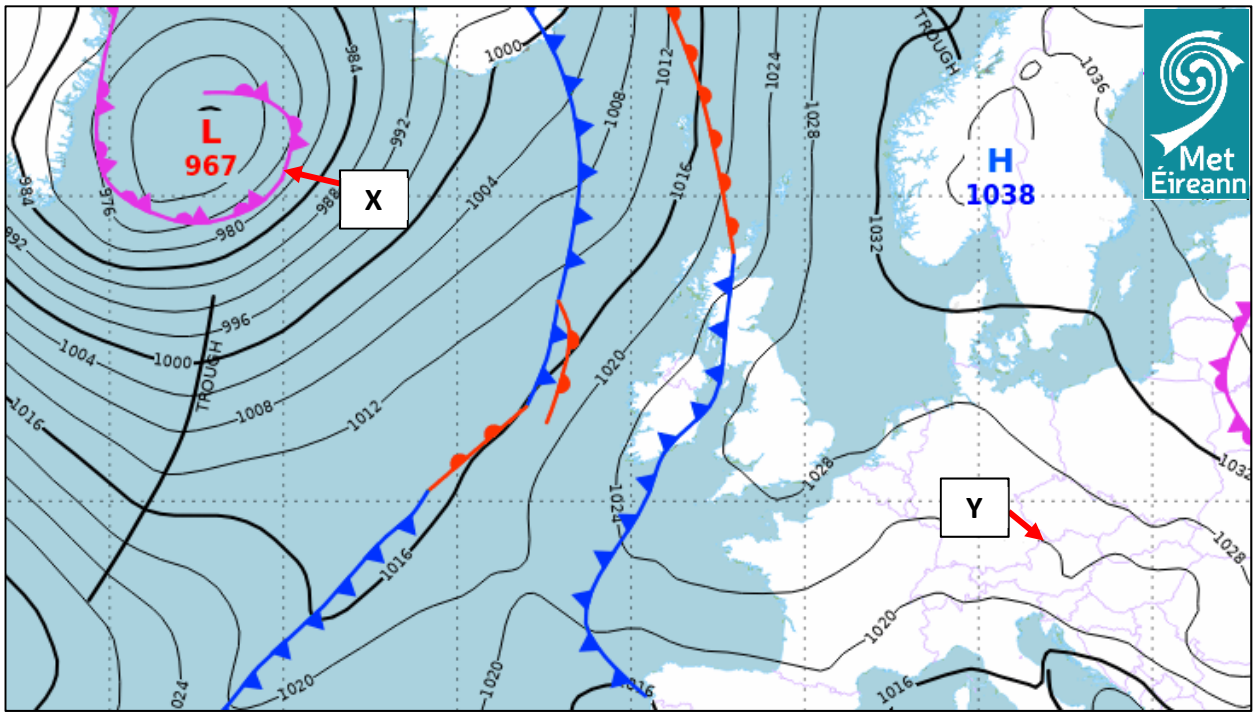
Amended from Google Streetview

Using accepted notation, give the location on the aerial photograph of each of the areas shown in the images **A**, **B**, **C** and **D**, in the table below.

Google Street View Image	Location on the Aerial Photograph
A	
B	
C	
D	

[8m]

10. Weather



Examine the weather chart above from Met Éireann and answer each of the following questions.

(i) Which of the following types of front is labelled **X** on the weather chart above? Tick (✓) the correct box.

Warm front ☐

Cold front ☐

Occluded front ☐

(ii) What is the barometric pressure (in hectopascals/millibars) at **Y** on the weather chart?

(iii) Indicate whether the following statement is true or false by ticking (✓) the correct box.

Winds over the area marked **X** are stronger than the winds over the area marked **Y.**

True ☐

False ☐

(iv) What is the term used to describe the wind that blows most frequently over an area?

[8m]

11. Statistical Interpretation

Towns with the Highest Percentage of Daily Irish Speakers, 2016

Town	County	% Daily Irish Speakers	Number of Daily Irish Speakers	Total Number of People
Mín Lárach	Donegal	73	**X**	400
Rann na Feirste	Donegal	67	202	302
An Cheathrú Rua	Galway	62	467	753
Bun na Leaca	Donegal	58	223	385
An Bun Beag-Doirí Beaga	Donegal	53	766	1,445

Amended from CSO

Examine the table above showing information on towns with the highest percentage of daily Irish speakers and answer each of the following questions.

(i) Which town from the table above is in Co. Galway?

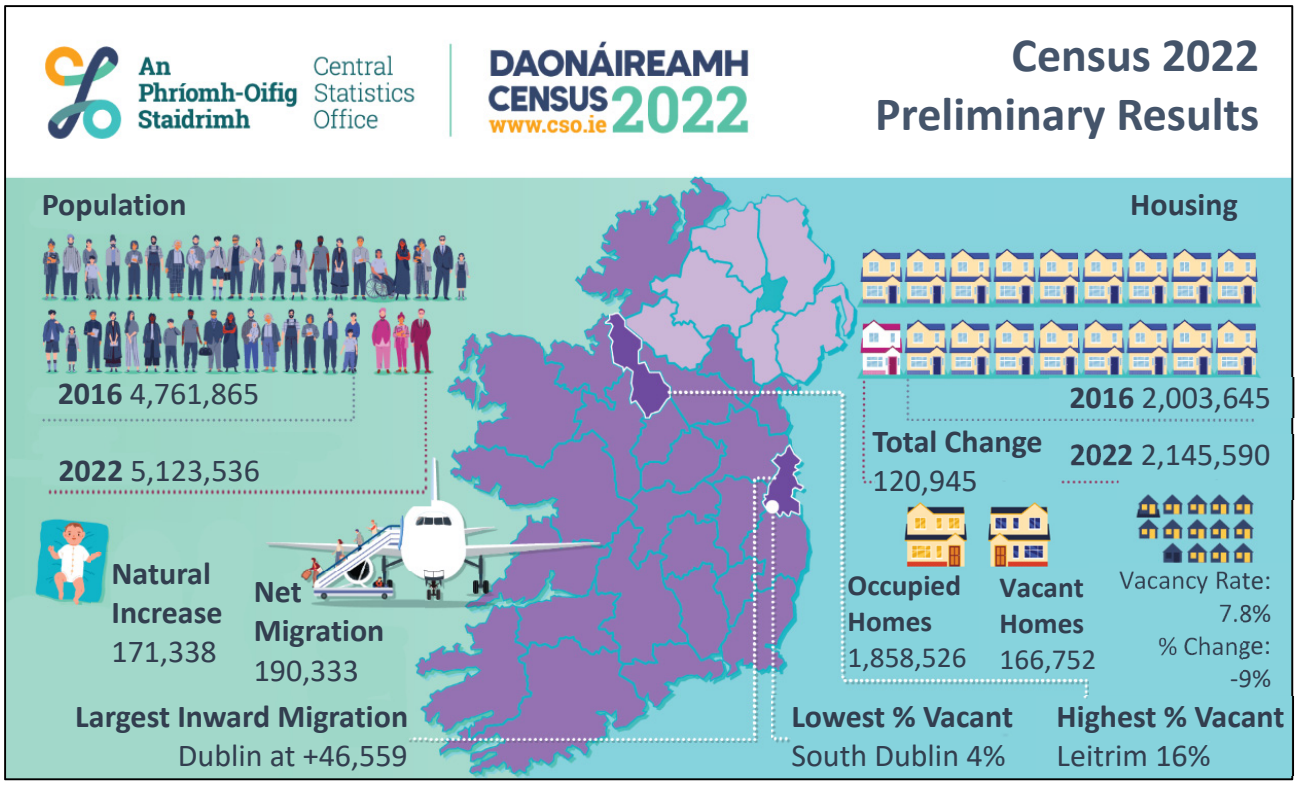
(ii) What was the percentage of daily Irish speakers in An Bun Beag-Doirí Beaga?

(iii) Which town from the table above had the lowest number of people in 2016?

(iv) Calculate **X** the number of people living in Mín Lárach in 2016 who were daily Irish speakers.

[8m]

12. Graphical Interpretation



Examine the infographic above on preliminary results of Census 2022 and answer each of the following questions.

(i) How many homes in Ireland were vacant in 2022?

(ii) Which county had the largest inward migration in 2022?

(iii) By how many people did the population of Ireland increase between 2016 and 2022?

(iv) State **two** ways that census data can be used by governments.

1.
2.

[8m]

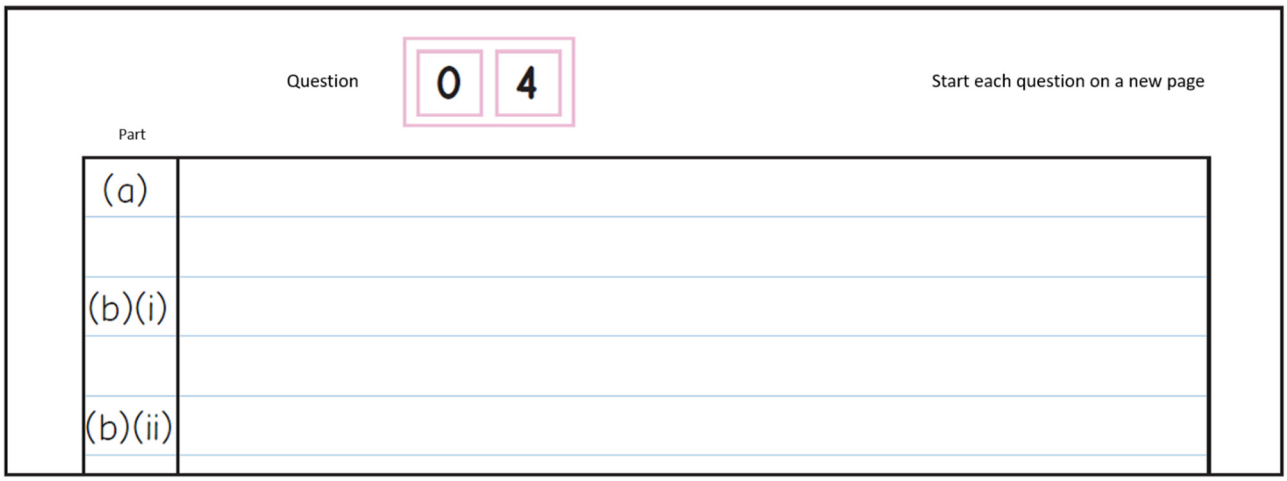
Answerbook for PART TWO

Instructions

Questions for **PART TWO** are supplied separately.

Start each question on a new page. Write the question number in the box at the top of each page.

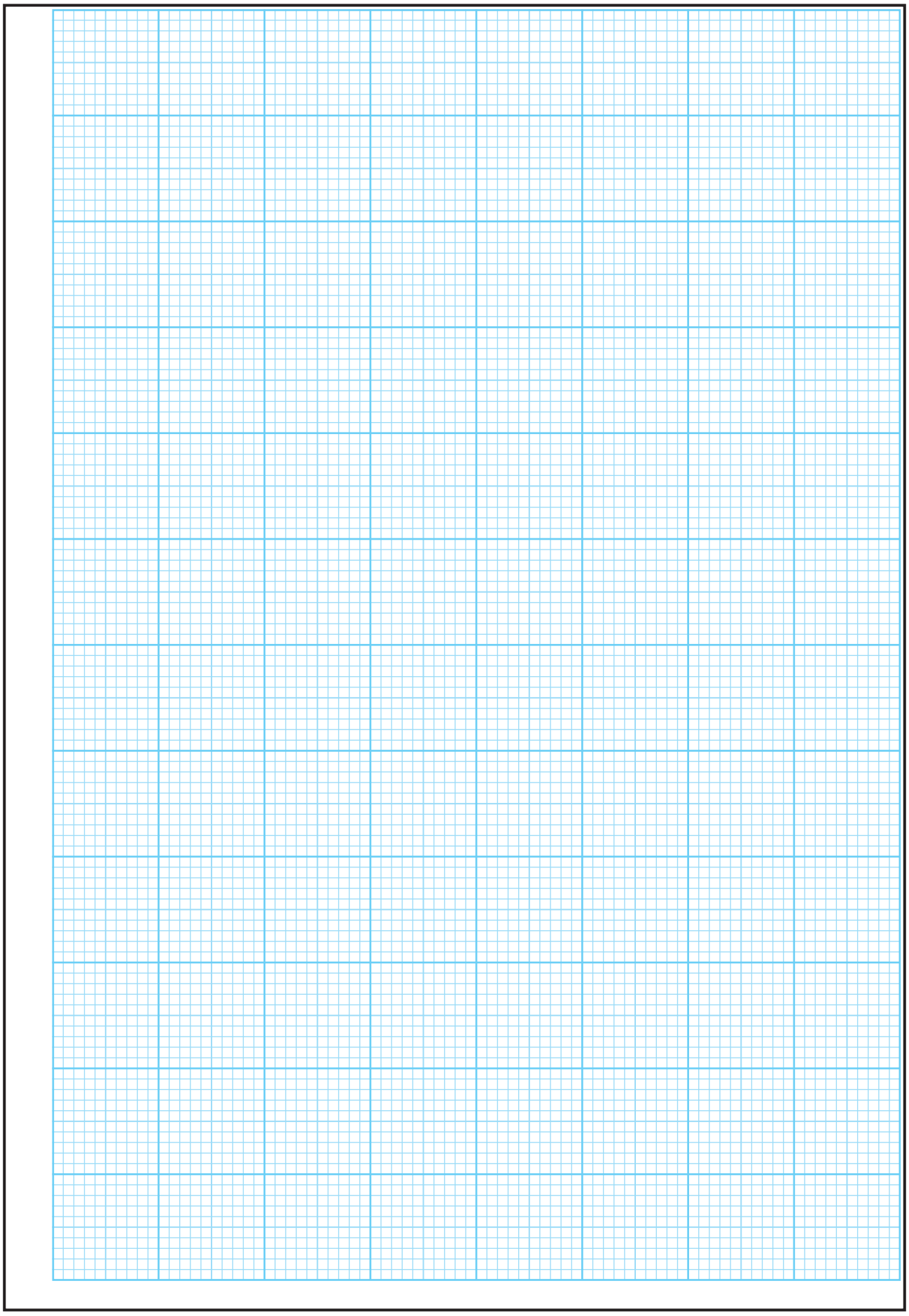
Use the left-hand column to label each part, as shown below.



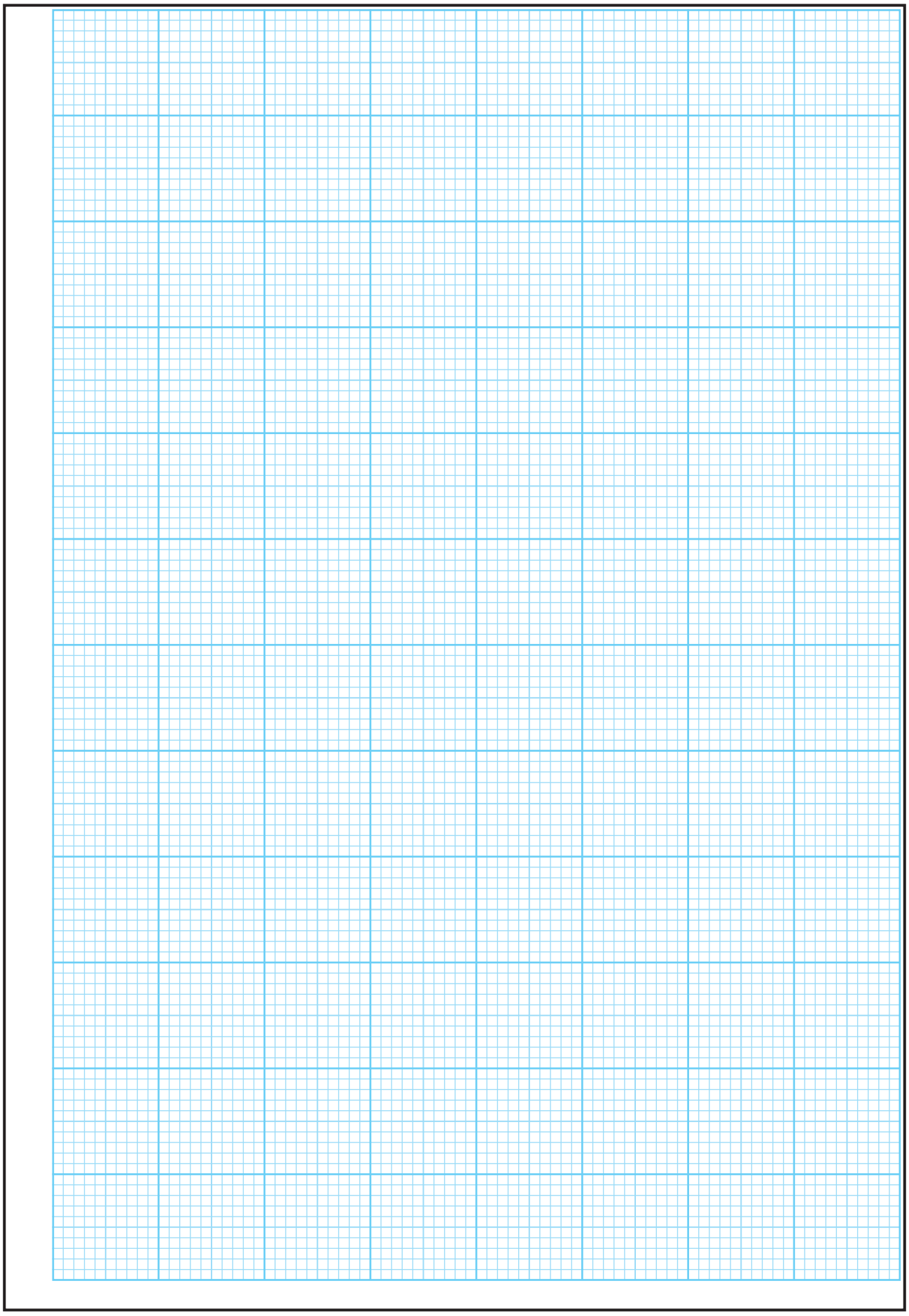
Four pages of graph paper are provided on the next four pages of this answerbook. On pages with graph paper, the box for the question number is at the bottom of the page.

You do not need to use all of the pages in this answerbook. If you run out of space in this answerbook, you may ask the superintendent for more paper or graph paper.

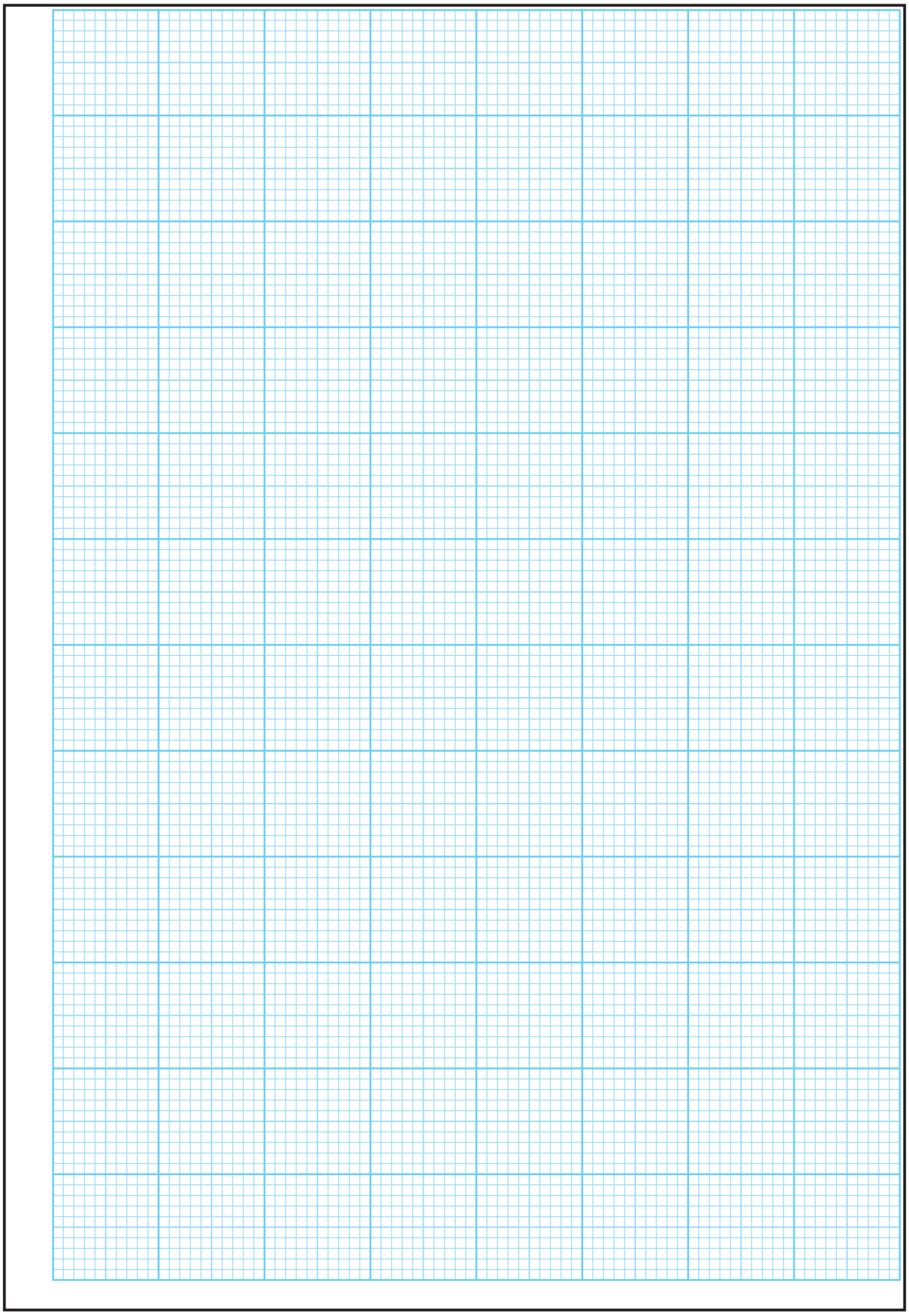
Write your answers in blue or black pen. You may use pencil for sketches, graphs and diagrams only.



Question



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Leaving Certificate – Higher Level

Geography Part One and Answerbook

Friday 9 June

Morning 9:30 - 12:20

Coimisiún na Scrúduithe Stáit
State Examinations Commission

Leaving Certificate Examination 2023

Geography

Part Two

Higher Level

Friday 9 June Morning 9:30 – 12:20

400 marks

Do not hand this up.

This document will not be returned to the State Examinations Commission.

PART TWO – STRUCTURED and ESSAY QUESTIONS – 320 MARKS

You must attempt **FOUR** questions. This must include one question from three of the sections below:

- Section 1: **Patterns and Processes in the Physical Environment** (Questions 1-3)
- Section 2: **Regional Geography** (Questions 4-6)
- Section 3: **Electives** (Questions 7-12)
- Section 4: **Options** (Questions 13-24)

The fourth question can be from any section.

All questions carry equal marks.

THESE QUESTIONS ARE TO BE ANSWERED IN THE ANSWERBOOK CONTAINING PART ONE

SECTION 1

PATTERNS AND PROCESSES IN THE PHYSICAL ENVIRONMENT

Questions 1 to 3

All questions carry 80 marks.

Question 1

1A. Ordnance Survey Map

Examine the 1:50 000 Ordnance Survey map and legend accompanying this paper.

Draw a sketch map of the area shown to half scale.

On it, correctly show and label each of the following:

- The entire area of Lisnananagh Lough
- The course of the Stradone River from where it exits Lisnananagh Lough to H 505 056.
- The triangulation pillar at Fleming's Folly in the south west of the map
- The entire area of land above 300 metres.

[20m]

1B. Igneous Rocks

Explain the formation of **two** igneous rocks, with reference to Irish examples.

[30m]

1C. Isostasy

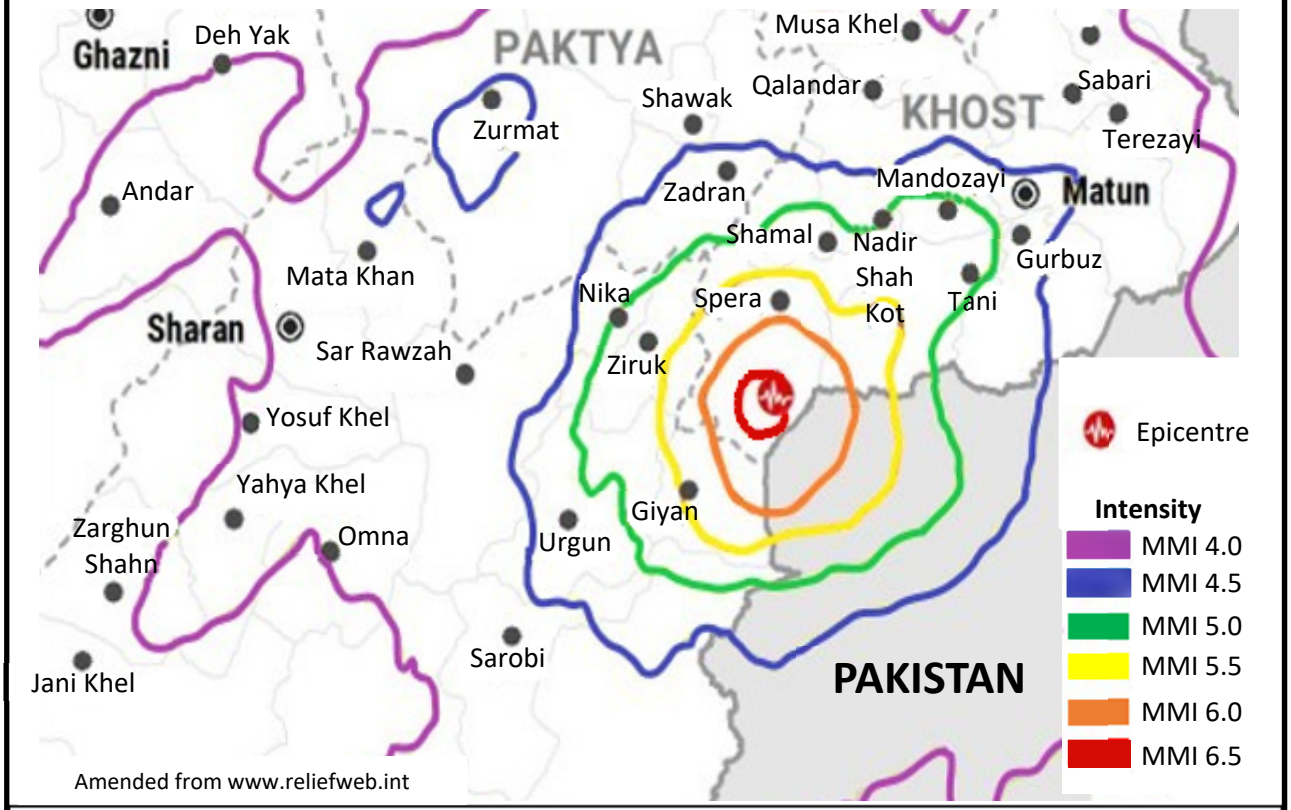
Examine how isostasy has impacted on the Irish landscape, with reference to the fluvial landscape and/or the coastal landscape.

[30m]

Question 2

2A. Earthquakes

Afghanistan Earthquake 22nd June 2022



Modified Mercalli Scale Intensity (MMI)				
2-3: Weak	4: Light	5: Moderate	6: Strong	7: Severe

Examine the map and legend above which gives information on an earthquake in Afghanistan and answer each of the following questions.

(i) What was the Modified Mercalli Scale Intensity (MMI) at the epicentre of this earthquake?

(ii) Name **two** settlements that experienced an MMI of 5.0 or higher.

(iii) On Afghanistan's border with which other country did this earthquake take place?

(iv) Other than the Modified Mercalli Scale, name **one** scale used to measure earthquakes.

(v) Name **one** instrument used to measure earthquakes.

(vi) Explain briefly the term *focus*, in relation to earthquakes.

(vii) Explain briefly **one** way to reduce the effects of an earthquake.

[20m]

2B. Surface Processes

Answer (i) **or** (ii)

(i) Examine the role of the **process of deposition** on the formation of **one** fluvial landform **or one** coastal landform **or one** glacial landform that you have studied.

Or

(ii) Describe and explain the **factors governing the operation** of **one** mass movement process that you have studied.

[30m]

2C. Chemical Weathering

Explain, with the aid of a labelled diagram(s), how chemical weathering has shaped **one** of the following:

- The surface karst landscape
- The underground karst landscape.

[30m]

Question 3

3A. Mass Movement

Glacier Avalanche in Italy

A mass of ice broke off the Marmolada glacier, sending an avalanche of ice, rocks and debris plummeting down the slope and onto a popular hiking trail along the Dolomite Mountains. Known as the Queen of the Dolomites, the Marmolada is about 3,300 metres high and is the tallest peak in the eastern Dolomites. According to research by Italian scientists the Marmolada glacier has lost an estimated 80% of its volume over the last 72 years and there was a warning in 2020 that the glacier could disappear within 15 years because of global warming.



Amended from www.theguardian.com

Examine the article above which gives information on a glacier avalanche and answer each of the following questions.

(i) In which country are the Dolomite Mountains located?

(ii) What height (in metres) is the Marmolada mountain?

(iii) By what estimated percentage has the volume of the glacier been reduced in the last 72 years?

(iv) Name **one** example of a slow type of mass movement.

(v) Explain briefly the term *regolith.*

(vi) List **two** factors that influence the speed of mass movement.

(vii) Explain briefly how **one** of these factors influences the speed of mass movement.

[20m]

3B. The Tectonic Cycle

Explain how the study of plate tectonics has helped us to understand the global distribution of any **one** of the following:

- Earthquakes
- Volcanoes
- Fold mountains

[30m]

3C. Human Interaction with the Rock Cycle

Examine how humans interact with the rock cycle, with reference to **one** of the following:

- Mining
- Extraction of building materials
- Oil/gas exploitation
- Geothermal energy production.

[30m]

SECTION 2

REGIONAL GEOGRAPHY

Questions 4 to 6

All questions carry 80 marks.

Question 4

4A. Map Skills

Draw an outline map of a **European** region (not in Ireland) that you have studied.

On it, show and name each of the following:

- The outline of a named feature of relief (upland or lowland)
- A named urban centre
- The outline of a named agricultural or industrial region
- The course of a named river.

[20m]

4B. Tertiary Economic Activity – Continental / Sub-Continental Region

Account for the development of **one** tertiary economic activity in a **Continental / Sub-Continental** region (not in Europe), that you have studied.

[30m]

4C. Urban Development

Examine how **two** of the following factors have influenced the development of any urban area that you have studied:

- Transport
- Location
- Primary economic activity.

[30m]

Question 5

5A. European Air Transport

Goods Transported by Air (in thousands of tonnes)

	2015	2018	2021
Belgium	1,126	1,416	2,083
Denmark	210	242	246
France	2,380	2,407	2,275
Germany	4,325	4,773	5,273
Ireland	148	156	155
Netherlands	1,712	1,840	1,807
Portugal	133	173	174
Switzerland	401	493	341
EU 27	12,524	14,243	15,023

Amended from Eurostat

Examine the table above showing the goods transported by air in Europe in selected years and answer each of the following questions.

(i) Which country transported 242,000 tonnes of goods by air in 2018?

(ii) Which country had the lowest amount of goods transported by air in 2015?

(iii) Name **one** country that had a decline in goods transported by air between 2015 and 2021.

(iv) Calculate the percentage of EU 27 goods transported by air in 2018 that were accounted for by Germany.

(v) Which country from the table above is not a member of the EU?

(vi) Explain briefly **one** advantage of transporting goods by air.

(vii) Explain briefly **one** environmental challenge caused by air transport.

[20m]

5B. Primary Economic Activity

Explain any **two** factors that have influenced the development of **one** primary economic activity in an **Irish** region that you have studied.

[30m]

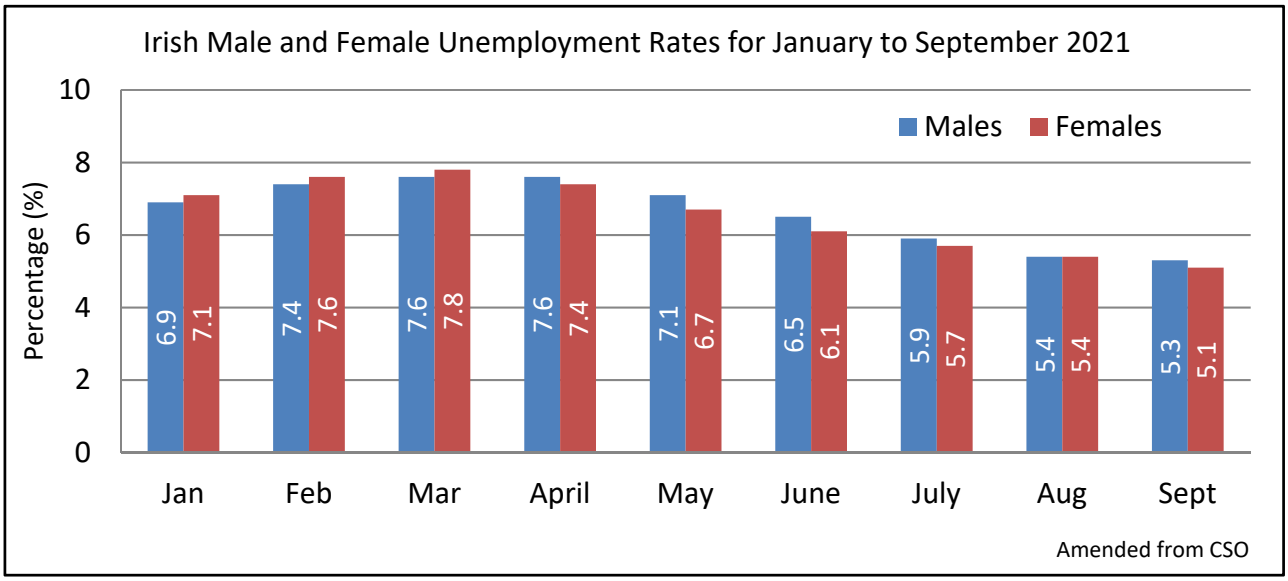
5C. The European Union

Discuss the potential challenges facing the future development of the European Union with reference to factors such as political union, economic union and sovereignty.

[30m]

Question 6

6A. Irish Unemployment



Examine the clustered bar chart above showing the male and female unemployment rates in Ireland from January 2021 to September 2021 and answer each of the following questions.

(i) What was the female unemployment rate (%) for March 2021?

(ii) In which month were the male and female unemployment rates the same?

(iii) For how many months was the female unemployment rate higher than the male unemployment rate?

(iv) Was the unemployment rate for males increasing **or** decreasing from May to September?

(v) Calculate the average unemployment rate (%) for females for the months shown on the chart above.

(vi) Explain briefly **two** benefits of having a low unemployment rate.

[20m]

6B. Secondary Economic Activity

Account for the development of secondary economic activity in a **European** region (not in Ireland) that you have studied, with reference to any **two** of the following factors:

- Labour
- Transport
- Raw materials
- Markets.

[30m]

6C. Population in Ireland

Account for the distribution of population throughout an **Irish** region that you have studied.

[30m]

SECTION 3

ELECTIVES

Questions 7 to 12

Patterns and Processes in Economic Activities (Questions 7, 8 and 9)

Patterns and Processes in the Human Environment (Questions 10, 11 and 12)

All questions carry 80 marks.

PATTERNS AND PROCESSES IN ECONOMIC ACTIVITIES

Questions 7 to 9

Question 7

7A. Aerial Photograph

Examine the aerial photograph accompanying this paper.

Draw a sketch map of the aerial photograph, half the length and half the breadth.

On it, correctly show and label each of the following:

- The lake
- **Two** connecting streets
- An area of waste ground in the centre foreground.
- Industrial buildings in the left background

[20m]

7B. Mobility of Economic Activities

Examine the mobility of modern economic activities with reference to **one** multinational company that you have studied.

[30m]

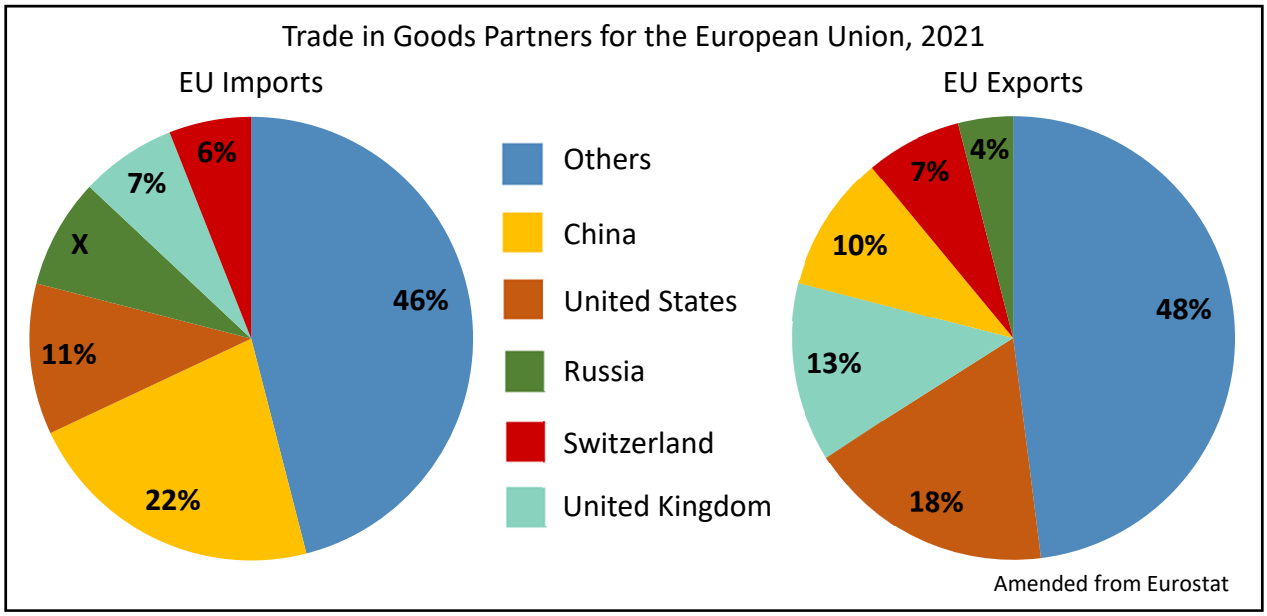
7C. Environmental Impact

Examine how economic activities can have an environmental impact with reference to examples on a local scale and a global scale.

[30m]

Question 8

8A. European Union Trade



Examine the pie charts above showing European Union (EU) trade in goods partners for 2021 and answer each of the following questions.

(i) Which country did the EU import 6% of all imported goods from?

(ii) What percentage of EU exports went to the United States?

(iii) What was the percentage difference in exports and imports between the EU and the United Kingdom in 2021?

(iv) Name **one** country shown above that the EU had a trade surplus with in 2021.

(v) Calculate **X**, the percentage of imports to the EU that came from Russia in 2021.

(vi) Explain briefly the term *balance of trade*.

(vii) Explain briefly **one** way that change in EU membership has had an impact on EU trade.

[20m]

8B. Economic Development – Developed Economy

Examine the development of **one** of the following with reference to a developed economy that you have studied:

- Footloose industries
- Financial services
- Mass tourism.

[30m]

8C. Economic Development

Using evidence from the 1:50 000 Ordnance Survey map that accompanies this paper, explain **three** factors that have influenced the development of economic activity in the area shown on the Ordnance Survey map.

[30m]

Question 9

9A. Economic Development

Average Total Years of Schooling for Adult Population for Selected Countries in 2017

Country	Years
Argentina	10
Burundi	3
Germany	14
Indonesia	8
Ireland	13
Niger	2

Amended from www.ourworldindata.org

Examine the data above showing the average total years of schooling for the adult population for selected countries in 2017 and answer each of the following questions.

(i) Using graph paper, draw a suitable graph to illustrate this data.

(ii) Explain briefly **one** disadvantage of using the Human Development Index (HDI) as a measure of economic development.

[20m]

9B. Colonialism

Examine the impact of colonialism on a developing economy that you have studied.

[30m]

9C. Renewable Energy

Examine how the development of renewable energy is crucial for Ireland's environment and economy.

[30m]

PATTERNS AND PROCESSES IN THE HUMAN ENVIRONMENT

Questions 10 to 12

Question 10

10A. Aerial Photograph

Examine the aerial photograph of Cavan accompanying this paper.

Draw a sketch map of the aerial photograph, half the length and half the breadth.

On it, correctly show and label each of the following:

- The lake
- An ecclesiastical land-use
- A carpark
- A residential land-use in the background.

[20m]

10B. Urban Issues

Discuss any **two** of the following issues that may arise from the growth of urban settlements:

- Urban decay and absence of community
- Urban sprawl
- Environmental issues.

[30m]

10C. Migration

Describe and explain changes in patterns of migration to and from Ireland since 1950.

[30m]

Question 11

11A. Asylum Applications in the EU

Asylum Applications to the European Union in 2021

Country	% of All Applications
Austria	6
France	19
Germany	30
Italy	8
Spain	10
Rest of EU	27

Amended from www.consilium.europa.eu

Examine the data above showing the percentage of all asylum applications to the EU received by different countries in 2021 and answer each of the following questions.

(i) Using graph paper, draw a suitable graph to illustrate this data.

(ii) Explain briefly **one** reason why people seek asylum.

[20m]

11B. Dynamics of Population

Examine the impact of any **one** of the following factors on levels of population change, with reference to example(s) that you have studied:

- The development of resources
- Income levels
- Technology.

[30m]

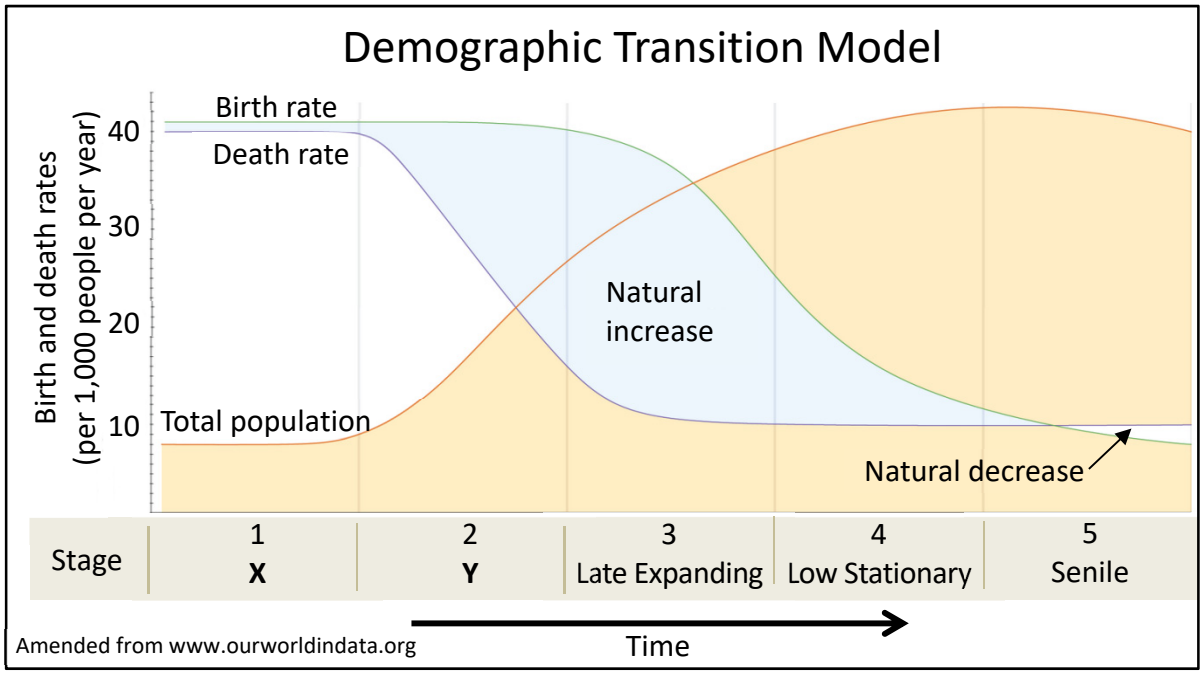
11C. Urban Land-Use

Describe and explain the land-use zones in any city that you have studied.

[30m]

Question 12

12A. Population Dynamics



Examine the diagram above showing the Demographic Transition Model and answer each of the following questions.

(i) Name **X**, Stage 1 of the Demographic Transition Model.

(ii) Name **Y**, Stage 2 of the Demographic Transition Model **and** name an example of a country in this stage.

(iii) What stage of the Demographic Transition Model is Ireland currently in?

(iv) In which stage of the Demographic Transition Model does the death rate reduce by the most?

(v) State **one** reason why the birth rate starts to decrease after stage 2.

(vi) Explain briefly **one** challenge facing countries in Stage 5 of the Demographic Transition Model.

(vii) Explain briefly the term *natural change.*

[20m]

12B. Rural Settlement Patterns

Using evidence from the 1:50 000 Ordnance Survey map of Cavan accompanying this paper, describe and explain **three** rural settlement patterns evident on the map.

[30m]

12C. Urban Growth – Developing World City

Problems can develop from the growth and expansion of urban centres.

Discuss this statement with reference to **one** developing world city that you have studied.

[30m]

SECTION 4

OPTIONS

Questions 13 to 24

All questions carry 80 marks.

Note: It is better to discuss **three** or **four** aspects of the theme in some detail, rather than to give a superficial treatment of a large number of points.

Global Interdependence

Note: It is better to discuss **three** or **four** aspects of the theme in some detail, rather than to give a superficial treatment of a large number of points.

13. Empowering people is a way of linking economic growth with human development. Discuss this statement with reference to any **two** of the following:

- Aid programmes
- Land ownership patterns
- Differing gender roles in society.

[80m]

14. Views of development and underdevelopment involve contrasting models and approaches to development.
Discuss.

[80m]

15. We live in an interdependent global economy. Discuss the impact of social and political decisions with reference to each of the following:

- Economic and political refugees
- Migration patterns
- Human rights issues.

[80m]

Geoecology

Note: It is better to discuss **three** or **four** aspects of the theme in some detail, rather than to give a superficial treatment of a large number of points.

16. Soil characteristics are affected by a combination of soil processes operating in their immediate environment. Examine how any **three** of the following processes impact on soil characteristics:

- Weathering
- Soil erosion
- Leaching
- Podzolisation.

[80m]

17. Biomes are altered by human activities. Discuss.

[80m]

18. Describe and explain the general composition and characteristics of any **one** soil type.

[80m]

Culture and Identity

Note: It is better to discuss **three** or **four** aspects of the theme in some detail, rather than to give a superficial treatment of a large number of points.

19. Many cultural groups around the world do not have a nation state of their own. Discuss this statement with reference to example(s) you have studied.

[80m]

20. Examine how people express their culture and identity in everyday life.

[80m]

21. Examine the strategies that have been or could be implemented to aid the survival of minority languages.

[80m]

The Atmosphere – Ocean Environment

Note: It is better to discuss **three** or **four** aspects of the theme in some detail, rather than to give a superficial treatment of a large number of points.

22. Examine how solar energy is transformed and redistributed through circulation patterns in the atmosphere and oceans.

[80m]

23. Give an account of how atmosphere-ocean phenomena (e.g. pressure, temperature, wind and humidity) are measured and examine the importance of such measurements.

[80m]

24. Examine how climate characteristics influence economic development.

[80m]

Do not hand this up

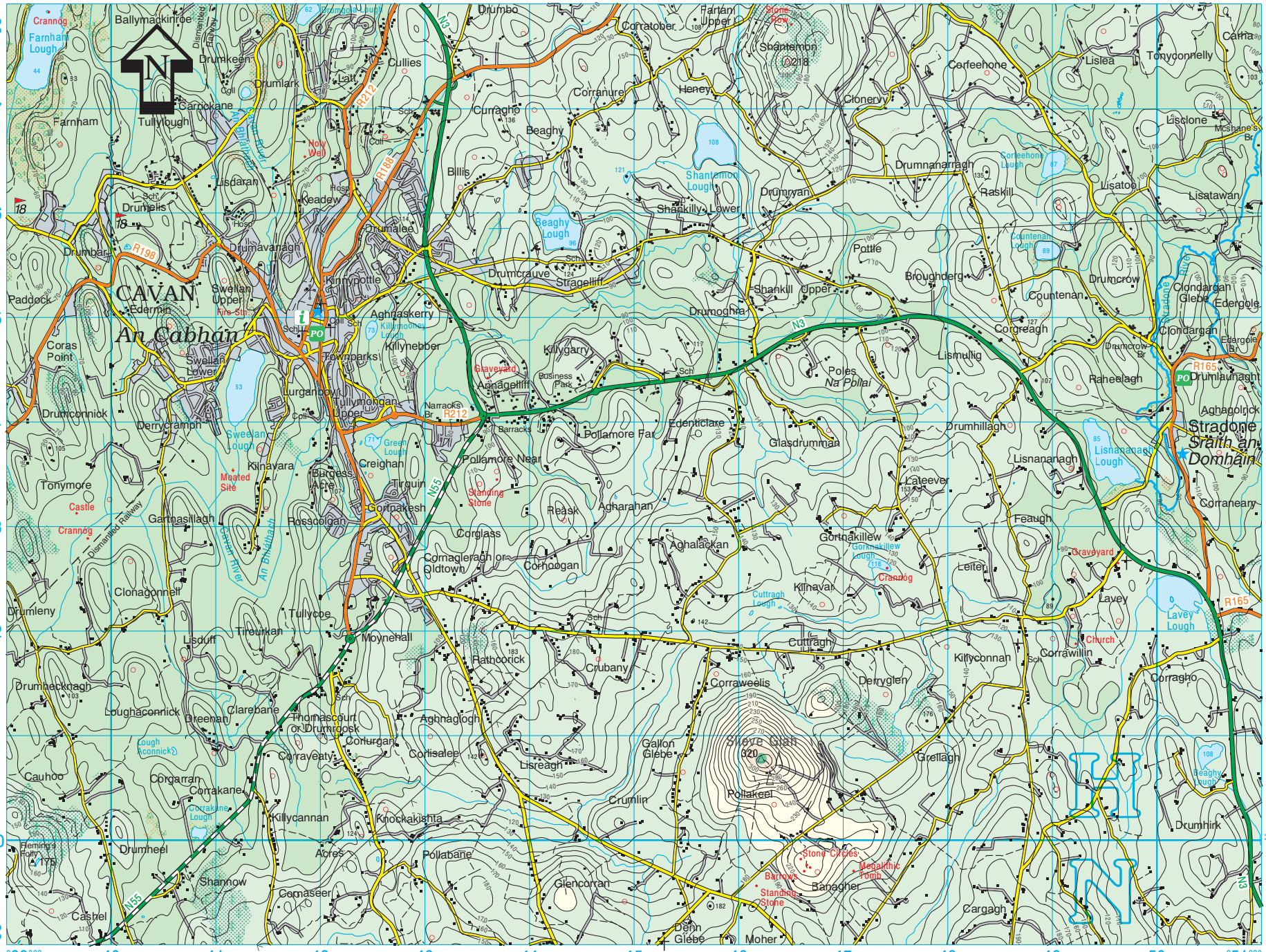
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Leaving Certificate – Higher Level

Geography Part Two

Friday 9 June
Morning 9:30 – 12.20



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Legend
Eochair

Ordnance Survey Ireland
Suirbhéireacht Ordanáis Éireann

DISCOVERY SERIES
SRAITH EOLAIS

M 1 — 1 — Mótarbhealach
Motorway (Junction number)

N 11 — Bóthar príomha náisiúnta
National Primary Road

N 71 — Bóthar tánaisteach náisiúnta
National Secondary Road

Carrbhealach dúbailte
Dual Carriageway

Bóthar príomha /tánaisteach náisiúnta beartaithe
Proposed Nat. Primary / Secondary Road

R 574 — Bóthar Réigiúnach
Regional Road

4 metres min / 4 metres max — Bóthar den tríú grád
Third Class Road

Boithre de chineál eile
Other Roads

Bealach
Track

Líne tarchurtha leictreachais
Electricity Transmission Line

Stáisiún cumhachta (uisce)
Power Station (Hydro)

Stáisiún cumhachta (breosla iontaiseach)
Power Station (Fossil)

Crann
Mast

Brú de chuid An Óige
Youth Hostel (An Óige)

Brú saoire Neamhspleách
Independent Holiday Hostel

Láithreán carbhán (idirthurais)
Caravan site (transit)

Láithreán campála
Camping site

Láithreán picnicí
Picnic site

Ionad dearctha
Viewpoint

Ionad pairceála
Parking

A T — An Taisce
National Trust

Tearmann Dúlra
Nature Reserve

Feirm Ghaoithe
Wind Farm

Foirgnimh le hais a chéile
Built up Area

Ionad eolais turasóireachta (ar oscailt ar feadh na bliana)
Tourist Information centre (regular opening)

Ionad eolais turasóireachta (ar oscailt le linn an tséasúir)
Tourist Information centre (restricted opening)

Garda Síochána
Police

PO — Oifig phoist
Post office

Eaglais no séipéal
Church or Chapel

Ardeaglais
Cathedral

Aerfort
Airport

Aerpháirc
Airfield

9 18 27 — Galfchúrsa, machaire gailf
Golf Course or Links

Bealach rothar
Cycle route

Siúlbhealach le comharthaí; Ceann Slí.
Waymarked Walks; Trailheads.

Loch
Lake

Canáil, canáil (thirim)
Canal, Canal (dry)

Abhainn nó sruthán
River or Stream

Líne bharr láin
High Water Mark

shingle,mud sand or loose rock — Líne lag trá
Low Water Mark

Trá
Beach

Ferry V — Bád fartha (feithiclí)
Ferry (Vehicle)

Ferry P — Bád fartha (paisinéirí)
Ferry (Passenger)

Teach Solais in úsáid / as úsáid
Lighthouse in use / disuse

Bádóireacht
Boating activities

Iarnróid
Railways

Iarnród tionscalaíoch
Industrial Line

Tollán
Tunnel

LC — Crosaire comhréidh
Level Crossing

Staisiún traenach
Railway Station

Teorainn idirnáisiúnta
International Boundary

Teorainn chontae
County Boundary

An Ghaeltacht
Irish speaking area

Páirc Náisiúnta
National Park

Páirc Foraoise
Forest Park

Seilbh de chuid an Aire Chosanta
Dept. of Defence Property

Foraois bhuaircíneach
Coniferous Plantation

Coilearnach Dhuillsilteach
Deciduous Woodland

Foraois mheasctha
Mixed Woodland

Séadchomhartha Ainmnithe
Named Antiquities

Clós, m.sh. Ráth nó Lios
Enclosure, e.g. Ringfort

Láthair Chatha (le dáta)
Battlefield (with date)

This is a sample reference only

(Discovery Sheet 23)

Sample reference: G 103 079

SUMMIT INFORMATION

Above 600m
599m - 400m
Below 400m

NOTE Over 600m summits must have a prominence of 15m
Between 400m and 599m a prominence of 30m and from 150 to 399m a prominence of 150m

The summit classification is courtesy the Mountain Views hillwalking community.
The lists used, updated to 2009, include:
The "Arderins" 500m list.
The "Vandeleur-Lynam" 600m list,
and other lists for smaller tops and county high points.

⊕ Mountain Rescue Base

1000 900 800 700 600 500 400 300 200 100

Céim imlíne comhairde 10m
10m Contour Interval

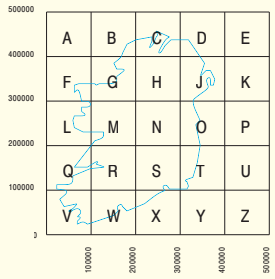
Céim imlíne comhairde 50m
50m Contour Interval

△ Cuaille triantánachta
Triangulation Pillar

123 • Spota airde
Spot Height

Trasnú cliathráin
Graticule Intersection

IRISH NATIONAL GRID



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Irish Transverse Mercator **Not used on this extract.**
(ITM) is a newly derived GPS compatible mapping projection that is associated with the European Terrestrial Reference System 1989 (ETRS89). For further information on ITM and for coordinate conversion visit our website.

CENTRE OF SHEET ITM CO-ORDINATES:
EXAMPLE: 499973E 827008N

SCALE 1:50 000
SCÁLA 1:50 000

1 KILOMETRES 0 1 2 3 4 5 6 7 KILOMETRES 8
1 STATUTE MILES 0 1 2 3 4 STATUTE MILES 5

2 ceintiméadar sa chiliméadar (taobh chearnóg eangaí) 2 centimetres to 1 Kilometre (grid square side)